

SMART INDIA HACKATHON 2026



Problem Statement ID: SIH26166

Problem Statement Title: Multimodal, Sun angle and
scale invariant image correspondence using Chandrayaan-2
optical images (OHRC, TMC and IIRS)

Theme: Space Technology

PS Category: Software

Team ID: 180758

Team Name: Severed Department



Chandralipi

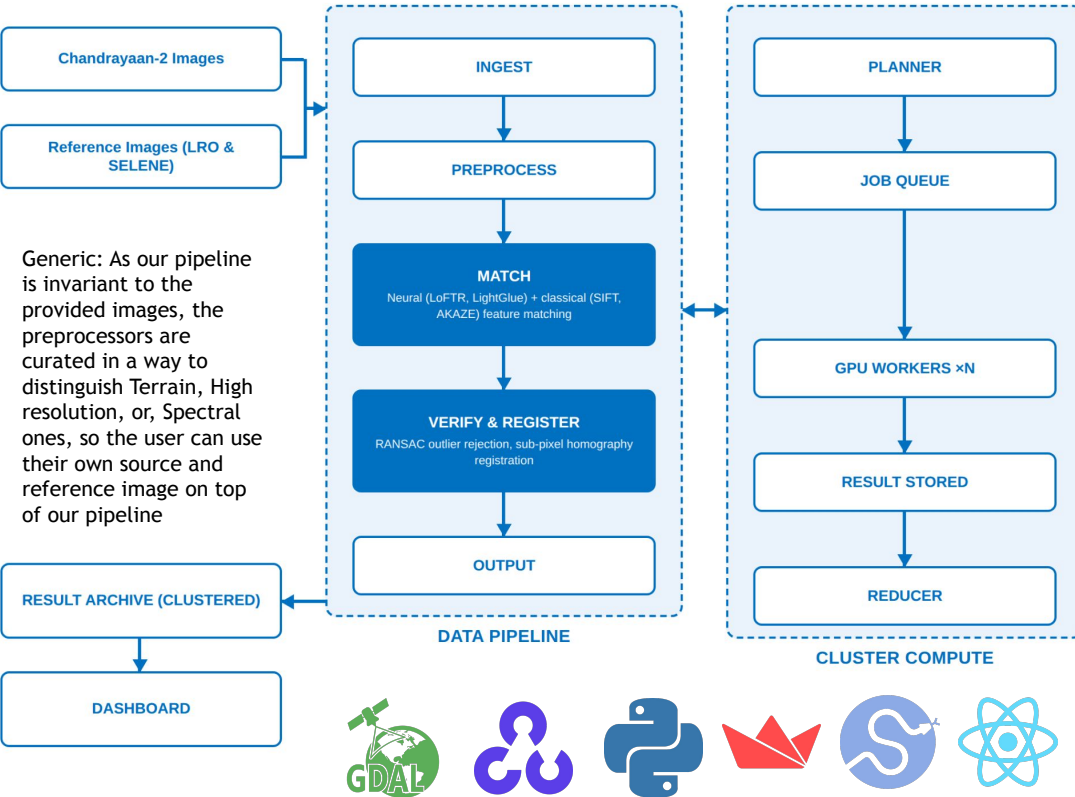
- **Automated Multi-Sensor Matching:** Chandralipi takes images from Chandrayaan-2's OHRC, IIRS and TMC-2 cameras and searches for matching points against reference images from NASA's LRO and JAXA's SELENE.
- **Illumination and Scale robust pipeline:** Chandralipi reconciles resolution gap, angle and illumination differences between the images from a wide specifications of cameras to recognize the same lunar feature.
- **Subpixel Accuracy:** It delivers precise uniformly spread correspondence points and reports spatial uniformity score instead of relying on visual checks from humans.
- **Coarse stage fine tuning:** Low rank tuning of matcher's coarse attention layer, to reduce false matches on repetitive of look alike crater terrain, on weak pseudo ground truth.
- **Cross modal embeddings for IIRS:** Bridges the optical and hyperspectral gap for IIRS.
- **Easy inclusive results Dashboard:** Chandralipi's result browsable interface showing every registered image, correspondence overlays and accuracy metrics.
- **Distributed Cluster Architecture:** Chandralipi is built on a work-queue system consisting of 3 stateless GPUs which can be easily scaled from one laptop to a huge computing cluster.
- **Problem Solved:** Registers and aligns OHRC/IIRS/TMC-2 multi-modal correspondence at subpixel accuracy with reference images of LRO, SELENE and also with each other with minimal manual involvement along with statistical metrics like RMSE, inlier count and ration as deliverables.

Novelty:

1. We are not only matching Chandrayaan-2 source images to Lunar Reference images but we are also registering them. IIRS is the hardest, for that we have build a pipeline starting from matching IIRS to Kaguya and SELENE multiband imager which is then registered with TMC-2 which is further registered to downsampled OHRC which can easily be registered to the original OHRC.
(IIRS <-> JAXA ref. <-> TMC-2 <-> OHRC)
2. We are also applying the concepts of Cluster Computing so that our software can be easily be deployed into other distributed systems with minimal effort and be scalable for organisations or small groups.
3. We are developing our own methodology with respect to previously made ones to find a better matching algorithm.



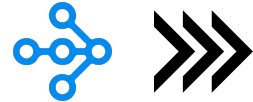
Technical Approach



- Core: Python.
- CV/Matching: SIFT, ASIFT, AKAZE, KAZE, ORB, BRISK, RIFT2 (custom), LoFTR, LightGlue, SuperGlue.
- AI/ML: PyTorch.
- Geospatial/Data: rasterio, Parquet, PDS4 ingest.
- Demo: Streamlit, React.
- Testing: pytest, ruff.
- Distributed: Ray, Docker, MinIO.
- Version control: GitHub.

Currently, we have successfully run the pipeline on two heterogeneous machines with OHRC and LRO NAC datasets, which are shown in the demo video.

We have created our own successful model similar to lightglue which for now provides decent results on the previously mentioned datasets.



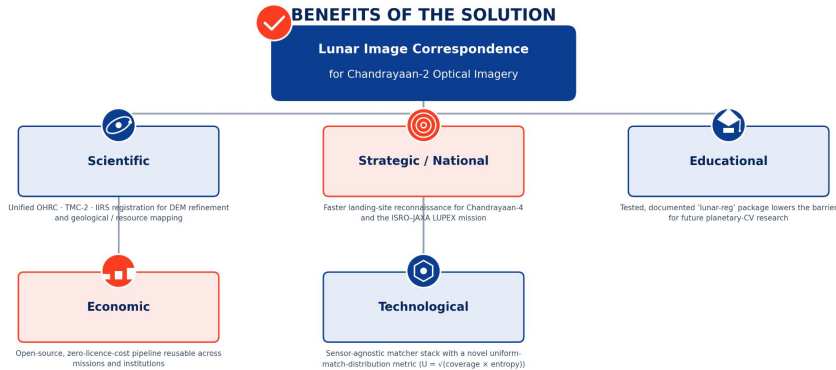


FEASIBILITY AND VIABILITY



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- **Technical Feasibility:** Proven matching techniques like SIFT, KAZE for classical ones and learned methods like LoFTR, LightGlue and license-gated SuperGlue, Chandralipi builds on top of them, along with that we are testing our in-house learned method; Easy registration methods and clear statistical results; Available cluster computing techniques.
- **Operational Feasibility:** Works on a small set of clustered GPUs in same manner as on a single GPU using work-queue of stateless GPU workers; Highly scalable and robust.
- **Key Challenges and Risks:** Disjoint lunar regions devoid of cross-instrumental overlap; Algorithm, like SuperGlue, licensing.
- **Strategies to Overcome Challenges:** LightGlue instead of SuperGlue due to licensing issues; PRADAN re-download filtered by shared footprint.
- **Economic and Scientific Viability:** Open Source data and open source tooling; Better detection of match failures using spatial uniformity metric instead of RMSE or manual efforts.



- **Economic:** Since it is open source, it reduces the cost of acquiring proprietary registration software for ISRO or any academic group working on Chandrayaan-2 or other comparable planetary archives.

- **Scientific:** Removes manual headache of adjusting all imagery into one coordinate frame, hereby supporting downstream Lunar Sciences like DEM refinement, mineral and resources characterization and so on.
- **National:** Chandralipi will shorten the reconnaissance cycle for future site selection and would turn ISRO's Chandrayaan-2 into a more usable resource.
- **Technological:** It can score match distribution explicitly unlike classical matchers where texture is strong, here the degradation is graceful as a generic sensor agnostic registration pipeline is used.
- **Educational:** It will remove barriers for students and researchers who would like to build planetary image correspondence work on top of it.



RESEARCH AND REFERENCES



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